



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

ASSIGNMENT-2

School of Engineering and Technology

Course Code & Name: ENCS304 Computer Networks

Program Name: B.Tech CSE

Name: Amandeep
Roll Number: 2301010200
Course: BTech CSE – C

Packet Flow Visualization Using Simulation Mode

Experiment Objectives:

Students will use Packet Tracer Simulation Mode to observe how packets move between devices, how switches learn MAC addresses, and how ICMP echo request/reply travels through a small network.

Learning Outcomes

- Use Simulation Mode effectively
- Observe ARP + ICMP behavior
- Understand switch MAC learning behavior
- Interpret packet event list timeline

Concepts Used

- Simulation Mode
- ICMP ping
- ARP resolution
- Switch MAC address table learning

Objective

The objective of this experiment is to understand how packets travel across a network using Simulation Mode in Cisco Packet Tracer. It also demonstrates ARP resolution, ICMP communication, and how switches learn MAC addresses.

Task 1: Building a Simple LAN

A simple Local Area Network (LAN) was created using:

- 1 Switch
- 3 PCs

IP Configuration

- PC1 → 192.168.20.1 /24
- PC2 → 192.168.20.2 /24
- PC3 → 192.168.20.3 /24

All devices were connected to the switch using straight-through cables.

Task 2: First Ping Observation (PC1 → PC2)

Simulation Mode was enabled and a ping was sent from PC1 to PC2.

Observed Behavior

1. ARP Request

- PC1 did not know the MAC address of PC2.
- It sent an ARP Request (broadcast) to all devices in the network.
- The switch forwarded this broadcast frame to all ports except the source port.

2. ARP Reply

- PC2 responded with its MAC address using an ARP Reply (unicast) to PC1.
- The switch forwarded this reply directly to PC1.

3. ICMP Echo Request

- After ARP resolution, PC1 sent an ICMP Echo Request (ping) to PC2.

4. ICMP Echo Reply

- PC2 responded with an ICMP Echo Reply.
- Communication was successfully completed.

Switch Behavior

- The switch learned MAC addresses by observing incoming frames.
- It stored:
 - PC1 MAC → Port X
 - PC2 MAC → Port Y

Task 3: Second Ping Observation (PC1 → PC2)

A second ping was sent from PC1 to PC2.

Observed Behavior

- No ARP request was generated.
- Only:

- ICMP Echo Request
- ICMP Echo Reply

Reason

- PC1 already had PC2's MAC address stored in its ARP cache.
- The switch also had MAC entries in its MAC address table.

Conclusion

- The second ping was faster and involved fewer packets.
-

Task 4: MAC Address Table Observation

The switch MAC address table was checked (or inferred through simulation events).

Observations

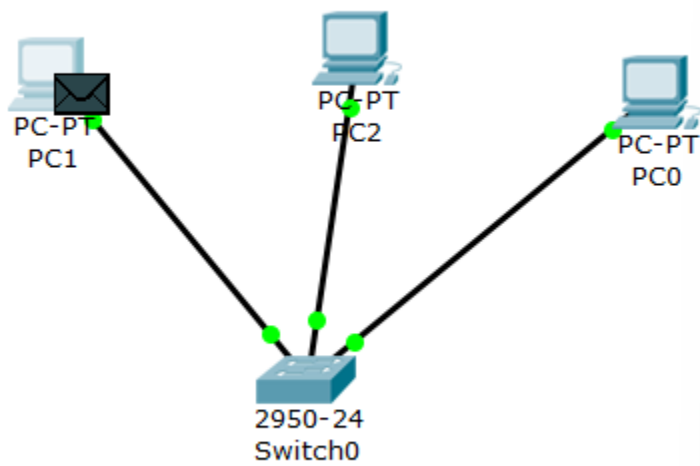
- The switch dynamically learned MAC addresses of connected PCs.
- Entries included:
 - MAC address of PC1 mapped to its port
 - MAC address of PC2 mapped to its port

Learning Mechanism

- When a frame arrives, the switch records:
 - Source MAC address
 - Incoming port
-

Screenshots

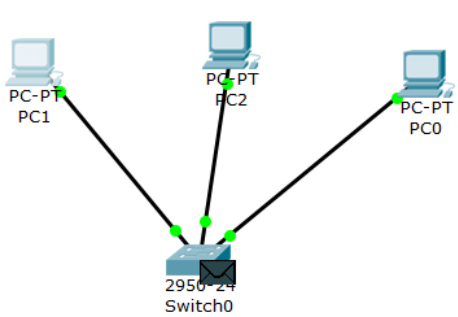
Setup



Ping

```
PC>ping 192.168.0.2  
  
Pinging 192.168.0.2 with 32 bytes of data:
```

transmission chronology



PC-PT
PC1

PC-PT
PC2

PC-PT
PC0

2950-24
Switch0

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devi	Type	Info
	0.000	--	PC1	ICMP	
	0.001	PC1	Switch0	ICMP	

Reset Simulation ☒ Constant Delay

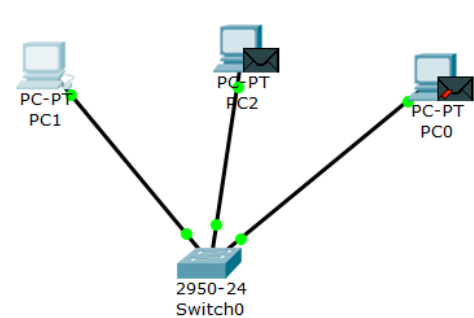
Play Controls

Back

[Root]

New Cluster

Move Object



PC-PT
PC1

PC-PT
PC2

PC-PT
PC0

2950-24
Switch0

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devi	Type	Info
	0.000	--	PC1	ICMP	
	0.001	PC1	Switch0	ICMP	
	0.002	Switch0	PC0	ICMP	
	0.002	Switch0	PC2	ICMP	

Reset Simulation ☒ Constant Delay

Play Controls

Back

[Root]
New Cluster
Move Object

PC-PT
PC1

PC-PT
PC2

PC-PT
PC0

2950-24
Switch0

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devi	Type	Info
	0.000	--	PC1	ICMP	
	0.001	PC1	Switch0	ICMP	
	0.002	Switch0	PC0	ICMP	
	0.002	Switch0	PC2	ICMP	
	0.003	PC2	Switch0	ICMP	

Reset Simulation ☒ Constant Delay

Play Controls

Back

[Root]
New Cluster
Move Object

PC-PT
PC1

PC-PT
PC2

PC-PT
PC0

2950-24
Switch0

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devi	Type	Info
	0.000	--	PC1	ICMP	
	0.001	PC1	Switch0	ICMP	
	0.002	Switch0	PC0	ICMP	
	0.002	Switch0	PC2	ICMP	
	0.003	PC2	Switch0	ICMP	
	0.004	Switch0	PC1	ICMP	

Reset Simulation ☒ Constant Delay

Play Controls

Back

Packet received

```
PC>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time=4ms TTL=128
```


Sending packet again after caching

```
PC>ping 192.168.0.2

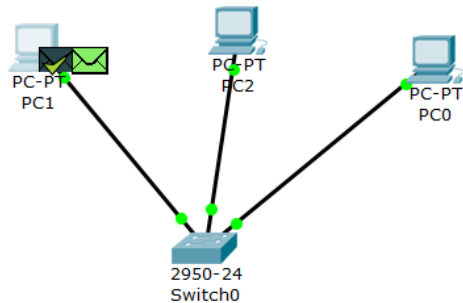
Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.0.2:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 4ms, Average = 4ms

Control-C
^C
PC>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:
```



Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devi	Type	Info
	0.000	--	PC1	ICMP	
	0.001	PC1	Switch0	ICMP	
	0.002	Switch0	PC0	ICMP	
	0.002	Switch0	PC2	ICMP	
	0.003	PC2	Switch0	ICMP	
	0.004	Switch0	PC1	ICMP	
	0.004	--	PC1	ICMP	

Reset Simulation

☒ Constant Delay

Play Controls

Back

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devic	Type	Info
	0.001	PC1	Switch0	ICMP	
	0.002	Switch0	PC0	ICMP	
	0.002	Switch0	PC2	ICMP	
	0.003	PC2	Switch0	ICMP	
	0.004	Switch0	PC1	ICMP	
	0.004	--	PC1	ICMP	
	0.005	PC1	Switch0	ICMP	

Reset Simulation
☒ Constant Delay

Play Controls

Back

[Root]
New Cluster
Move Object

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devic	Type	Info
	0.002	Switch0	PC0	ICMP	
	0.002	Switch0	PC2	ICMP	
	0.003	PC2	Switch0	ICMP	
	0.004	Switch0	PC1	ICMP	
	0.004	--	PC1	ICMP	
	0.005	PC1	Switch0	ICMP	
	0.006	Switch0	PC2	ICMP	

Reset Simulation
☒ Constant Delay

Play Controls

Back

[Root] New Cluster Move Object

PC-PT PC1 PC-PT PC2 PC-PT PC0

2950-24 Switch0

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devi	Type	Info
	0.002	Switch0	PC2	ICMP	
	0.003	PC2	Switch0	ICMP	
	0.004	Switch0	PC1	ICMP	
	0.004	--	PC1	ICMP	
	0.005	PC1	Switch0	ICMP	
	0.006	Switch0	PC2	ICMP	
	0.007	PC2	Switch0	ICMP	

Reset Simulation ☒ Constant Delay

Play Controls

Back

PC-PT PC1 PC-PT PC2 PC-PT PC0

2950-24 Switch0

Simulation Panel

Event List

Vis.	Time(sec)	Last Devi	At Devi	Type	Info
	0.003	PC2	Switch0	ICMP	
	0.004	Switch0	PC1	ICMP	
	0.004	--	PC1	ICMP	
	0.005	PC1	Switch0	ICMP	
	0.006	Switch0	PC2	ICMP	
	0.007	PC2	Switch0	ICMP	
	0.008	Switch0	PC1	ICMP	

Reset Simulation ☒ Constant Delay

Play Controls

Back

```
PC
PC>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time=4ms TTL=128
```

Packet received

Switch table (after caching)

```
Switch#show mac address-table
```

Mac Address Table

```
-----  
Vlan    Mac Address      Type      Ports  
----    -  
1       0090.0cc7.4992   DYNAMIC   Fa0/3  
1       00d0.ffbd.0c8a   DYNAMIC   Fa0/2  
Switch#
```
